

JUSTIFICATION AND APPROVAL FOR USE OF OTHER THAN FULL AND OPEN COMPETITION

1. Contracting Activity.

Contracting Activity: Naval Medical Logistics Command (NMLC), Acquisition Management Directorate, Code 02, 693 Neiman Street, Fort Detrick, MD 21702, is acquiring on behalf of the requesting agency: Naval Medical Research Command (NMRC), 503 Robert Grant Avenue, Silver Spring, MD 20901.

2. Description of the Action Being Approved.

Approval is requested to award a contract to pay the services provided by Mahidol University for endemic samples from dengue patients for a biomarker discovery study. The address of the contractor is below:

Mahidol University
999 Phutthamonthon 4 Rd., Salaya, Nakhon Pathom 73170, THAILAND

3. Description of Supplies/Services.

The contractor shall provide the following services including shipping and handling of the final products to NMRC:

3.1 Ship archived paired samples collected from 131 confirmed dengue cases between 2014-2016 to NMRC (68 uncomplicated, 63 moderate- severe dengue) by March/ April 2025 together with the related clinical and laboratory records.

3.2 Conduct prospective patient enrollment. The targeted enrollment number is 15 moderate-severe dengue and 40 uncomplicated dengue cases. The provider will collect serial samples at early acute (<96 hours from the fever onset), defervescence (<72 hours from the subsiding of fever), and convalescent (2-4 weeks after fever) from hospitalized patients confirmed with dengue infection. Samples are expected to be shipped to NMRC during March-April 2026.

3.3 The provider will provide demographic data and clinical characteristics including warning signs and results of baseline laboratory investigation such as CBC, AST, rapid NS1 tests for each case in 3.1 and 3.2.

The uniqueness of the service provider

Obtaining enough study subjects is the key for this proposal. Without the quantity and quality of the field samples, this study will fail.

In this study, the dengue cases will be grouped into three groups: dengue without warning sign (DENV-WS), dengue with warning sign (DENV+WS), and severe dengue (SD) based on the WHO 2009 Dengue Case Definition. Clinical endpoints such as plasma leakage, hemorrhage, and organ impairment (including neurological, hepatic, or cardiac involvement), will be also used for data analysis and marker discovery. These endpoints reflect the disease burden and severity, and the requirement of medical attention.

The goal of the study is to discover biomarkers associated with Severe Dengue or with study endpoints reflecting the disease burden and the requirement of intense medical care. Although dengue infection is

global, severe dengue is rare (<2%). Therefore, one of the biggest challenges of the study is the low occurrence of Severe Dengue. We have performed group size estimation using an online tool: Sample Size Calculator, using data from published papers. We found that a group size of 300 will likely ensure the statistical significance for 85-90% markers. Therefore, we set the goal of obtaining a subject group of 300.

The selection of the Thailand site is based on the epidemiology of dengue in the country. In the WHO Southeast Asia region, 10 out of 11 Member States are known to be endemic for dengue virus. Thailand ranked among the world's 30 most highly endemic countries for dengue. Compared to 2022, in 2023, Thailand recorded a larger number of dengue cases. By November 2023, Thailand saw an increase of over 300%, with dengue cases rising from 46 678 in 2022 to 136 655 in 2023 (as of 22 November 2023). During the same period, the death toll in Thailand increased from 34 (CFR 0.07%) to 147 (CFR 0.11%).

We discussed among the PIs of the field sites to increase the samples size. To achieve the goal of ~300 cases in the first 12 months of a 2-year funding period, the availability of archived samples becomes critical. The collaborator Dr. Yupin Suputtamongkol at Siriraj Hospital and Mahidol University (MU) has returned positive feedback as she has found 131 subjects with 3 time points of archived samples.

The Thailand site, led by Dr. Yupin Suputtamongkol, a professor and physician at Siriraj Hospital and Mahidol University, provides archived samples of interest. Dr. Yupin Suputtamongkol is a long-term collaborator of our department. She has archived samples from 2016. Active surveillance is year-round, capturing most current serotypes and strains.

Siriraj Hospital and Police General Hospital are clinical sites for enrolment of adult patients. The field team has a mature clinical laboratory where PCR or serology can be done to confirm the cases. This is also very key as many field sites lack resources and skilled workers for diagnostics capability.

They will collect acute and early convalescent samples. PBMCs will be collected if patients provide informed consent. For hospitalized patients, serial samples will be collected at early acute (first sample collected), <96 hours post-fever, and convalescent (2-4 weeks after fever). Demographic data and clinical characteristics including warning signs and results of baseline laboratory investigation such as CBC, AST, rapid NS1 test and outcome will be recorded in the standard case record form.

4. Statutory Authority Permitting Other Than Full and Open Competition.

10 U.S.C. 2304(c)(1), One source or limited sources.

5. Rationale Justifying Use of Cited Statutory Authority.

The contractor is the only one who has reliably responded to emails and expressed interest in providing the needed services with acceptable timeline for delivery.

6. Description of Efforts Made to Solicit Offers from as Many Offerors as Practicable.

Online for appropriate contractors:

Searching term: Commercial dengue patient samples

The search returned only one company [REDACTED]. We contact the company via email. The email conversation between NMRC and the company is enclosed. No positive response was received.

We therefore contacted our field collaborators who can perform prospective subject enrollment in Mahidol University, Thailand.

Dr. Yupin Suputtamongkol, a professor and physician at Siriraj Hospital and Mahidol University, Thailand. Dr. Suputtamongkol can provide archived samples and can run prospective enrollment. We selected her for the study site of Thailand.

We have no other collaborators in South-East Asia.

Our collaborator of NAMRU-SOUTH at Lima, Peru also has the capability of prospective enrollment of patients. However, there enroll patients outside of the hospitals, therefore will not have severe dengue cases.

7. Determination of Fair and Reasonable Cost.

Fair and reasonable determination will be made utilizing the Proposal analysis techniques described in FAR 15.404-1

8. Description of the market survey conducted and the results, or a statement of the reasons a market survey was not conducted.

Research was done by internet, emails, and TEAM meetings.

8. Other facts supporting the use of other than full and open competition.

If other suitable vendor(s) became available, a full and open competition may be considered.

9. Listing of Sources, if any, which expressed, in writing, an interest in the acquisition.

To date, no other sources have written to express an interest for a full service except the selected contractor, the Mahidol University.

11. Statement of actions, if any, the agency may take to remove or overcome any barriers to competition before any subsequent acquisition for the supplies or services required.

12. Point of Contact.

The point of contact at the Naval Medical Research Center, Silver Spring, MD, regarding the contents of the sole source justification is Shamai Carter who may be reached at (301) 319-7182.

**TECHNICAL AND REQUIREMENTS CERTIFICATION REQUIRED
BY NAVSUP 4200.83 series**

I certify that the facts and representations under my cognizance which are included in this justification and approval and which form a basis for this justification are complete and accurate.

Technical Cognizance

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Dr. Peifang Sun
PI
DSD/NMRC

Phone

Date

Requirements Cognizance

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Dr. Eric Hall
Department Head
DSD/NMRC

Phone

Date

CONTRACTING OFFICER CERTIFICATION

I certify that this justification is accurate and complete to the best of my knowledge.

CARTER.SHAMAI.

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Shamai M. Carter
Contracting Officer

(301) 319-7182

Phone

3/10/2025

Date